Checkpoint C1

Monday, October 26, 2026 4:48 PM

a) For Protocol A

G has some zeros and ones, added to A changes the main diagonal from all 0's to all 1's.

$$\begin{bmatrix} 1+d_1 & 0 & 0 \\ 0 & 1+d_2 & 0 \\ 0 & 0 & 1+d_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ \text{random } \{0,1\} & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$x_i[k+1] = \frac{(A+G)_i}{1+d_i} \; x[k]$$

- $x_i[k+1]$: robot i's new location
- $\frac{(A+G)_i}{1+d_i}$: total # of connections, including itself
- $x[k]$: entire list of x co-ords @ time step k
- $(A+G)_i$: filters only those robots that are in range

Protocol B

$L = \begin{bmatrix} d_1 & \text{random } 0\text{'s and } -1\text{'s} \\ \text{random } 0\text{'s and } -1\text{'s} & d_2 \\ & d_3 \end{bmatrix}$ — represents the size of a step.

For ϵL, ϵ divides each non-zero term by the max

$$x_i[k+1] = \begin{bmatrix} 1-\epsilon d_1 & 0 & 0 \\ 0 & 1-\epsilon d_2 & \epsilon \\ \epsilon & 0 & 1-\epsilon d_3 \end{bmatrix} x[k]$$

blue stuff could be anywhere except main diagonal

$$x_i[k+1] = (1-\epsilon d_i)(x_i[k]) + \sum \epsilon x_j[k]$$

- $(1-\epsilon d_i)$: how much robot "i" will stay i" place, if few neighbors then $1-\epsilon d_i$ close to 1
- $\sum \epsilon x_j[k]$: weighted average of neighbors positions, few connections means this value is low.
- whichever entries in x are considered neighbors

Protocol A

b) $\frac{(A+G)_i}{1+d_i} \cdot 1 = 1$

since every value in every row = 1 numerator doesn't filter anything out, results in length of the row

also results in length, since every robot is connected to every robot

ends up being $\frac{1}{1} \cdot 1$

If all the robots are at the same position, the average position is that position, thus none of the robots move

Protocol B

$$\begin{bmatrix} 1-\epsilon d_1 & 0 & \epsilon \\ \epsilon & 1-\epsilon d_2 & 0 \\ 0 & 0 & 1-\epsilon d_3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

let m = max connections

becomes d_i since every entry = 1

$x_j[k]$ represents all the robot's positions that are neighbors

$$(1-\epsilon d_i)(1) + \left(\epsilon \sum x_j[k]\right)(1) = 1$$

$$(1-\epsilon d_i)(1) + (\epsilon d_i)(1) = 1$$

$$1 - \frac{d_i}{m} + \frac{d_i}{m} = 1$$

$$1 = 1 \; \checkmark$$

c) Protocol A

$$(1+D)(1) = \begin{bmatrix} 1+d_1 & 0 & 0 \\ 0 & 1+d_2 & 0 \\ 0 & 0 & 1+d_3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+d_1 \\ 1+d_2 \\ 1+d_3 \end{bmatrix}$$

each entry = # of connections for that robot, including itself

$$\begin{bmatrix} 1+d_1 \end{bmatrix} \cdot \frac{(A+G)_i}{1+d_1} = 1+d_1$$

ends up being the # of connections of robot 1 (d_1) + itself (1)

$$\cancel{1+d_1} \cdot \frac{1+d_1}{\cancel{1+d_1}} = 1+d_1 \; \checkmark$$

of connections each robot has will not change as k ↑, since $(1+D)(1)$ does not change

Protocol B

$$1^T(I-\epsilon L) = 1^T$$

$$1^T - 1^T \epsilon L = 1^T$$

each row

$$1^T - \epsilon(1^T D - 1^T G) = 1^T$$

$$1^T - \epsilon(d - d) = 1^T$$

row vector

since $G = G^T$, sum of columns = sum of rows = d

$$1^T - 0 = 1^T \; \checkmark$$

has will not change as k ↑, since $(I+D)(I)$ does not change

8) With $\varepsilon > \frac{1}{\max d_i}$, the rows wouldn't sum to 1. Additionally, for the robots with connections close to max d_i, there is a chance that their "stay in place" weight could be negative. Hence, the robot would keep oscillating and never stay in one spot.

$\vec{1}^T - 0 = \vec{1}^T$ ✓

W_B stays constant, therefore the weighted averages for each robot and its neighbor's positions stay the same